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Estimation of critical conditions of polymers based on monitoring the polymer recovery

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ABSTRACT

Liquid chromatography at critical conditions (LCCC) is a very attractive chromatographic technique on the border between the size exclusion and liquid adsorption mode of the liquid chromatography. The strong interest in LCCC arises from the fact that it is well suited to analyze the block lengths in segmented copolymers or the heterogeneities with regard to end groups present, for example, in functionalized polymers e.g., telechelics. In this paper a new method for identification of the critical conditions of synthetic polymers is proposed, which requires only one polymer sample with higher molar mass. The method is based on monitoring the recovery of the polymer sample from a column. The composition of the mobile phase is modified until the polymer sample is fully recovered from the column. The corresponding composition of the mobile phase is composition corresponding to LCCC. This new method was applied for the determination of critical conditions for polyethylene, syndiotactic polypropylene and isotactic polypropylene. The results of the new method will be compared to those of classical approaches and advantages will be pointed out.

1. Introduction

The concept of liquid chromatography at critical conditions (LCCC) was for the first time proposed by Belenky et al. [1] and Entelis et al. [2]. The LCCC is an intermediate mode of chromatography between size exclusion chromatography (SEC) and liquid adsorption chromatography (LAC). In SEC mode polymers are separated according to their hydrodynamic volume in solution, while in LAC mode they are separated on the basis of their enthalpic interactions with the stationary phase in a mobile phase system. At LCCC the entropic effects of size exclusion and the enthalpic effects of adsorption compensate each other and macromolecules of a given polymer elute at the same elution volume from the stationary phase irrespective of their molar mass.

LCCC is a practically important chromatographic method, because it enables to achieve separations, which are not realizable with other chromatographic methods. For example, LCCC has been applied for separations of block copolymers [3–11], to determine the average molar mass of constituent blocks in di- and tri-block copolymer (type A–B and A–B–A) [12,13] as well as for selective separations of end-functionalized polymers according to their functional groups [2,3,14,15]. Several groups have described successful separations of macromolecules according to their architecture (for example, linear from star shaped or linear from rings) [16–19] or their tacticity [20–25] with LCCC.

In the majority of the papers, the critical conditions are determined by evaluating the dependence between the elution volumes and the molar mass of a series of samples for a polymer of interest (SEC-LAC-plots) [1–23]. Namely, at a specific composition of a mixed mobile phase and at a given temperature, the polymer samples with different molar mass elute at the same elution volume. This specific composition of the mobile phase is the critical point for the polymer at the applied temperature in this chromatographic system. This method of determining the critical point may be laborious due to the fact that a series of a polymer with different molar masses needs to be injected in series of a mobile phase with different composition. This may be a particular impediment due to the fact that the required series of polymers are not easily accessible.

Cools et al. [26] proposed a graphical extrapolation of the critical composition from the dependence between the elution volume at peak maximum [E_{pmax}] and the composition of a binary mobile phase. Cools' method also requires a significant number of measurements similar to the classical procedure (i.e., SEC-LAC plots) at different compositions of a selected mobile phase with analogous

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Table 1
Polymer samples used for the experiments.

Sample code	M_w [kg/mol]	Đ
it-PP^{348}	348.3	8.01
st-PP^{196}	196.8	2.44
PE^{181}	181	1.59

drawbacks. These are avoided by the method proposed by Brun et al. [27,28] and used by Bashir et al. [29], which requires a single polymer sample of high molar mass and a gradient pump to generate a linear gradient. The critical conditions are then determined as the composition of a mobile phase when the polymer is just desorbed and eluted from the column in the solvent gradient. E_{pmax} for a polymer sample of high molar mass is then used as a basis to refine the critical point for that polymer further using Bashir's method. Bashir's method requires two measurements i.e., one injection of the polymer sample in a solvent gradient and one experiment to determine the composition of the mobile phase in the detector used i.e., the delay volume.

Berek et al. pointed out a serious problem of limited recovery of the sample from the column, connected with LCCC in some polymer/solvent/sorbent systems. The sample recovery from the column rapidly decreases with increasing weight average molar mass and approaches zero for macromolecules between 100,000 and 500,000 g/mol molar mass [4,30–32]. The dependence of sample recovery on its molar mass also means that the high molar mass fractions of a broadly distributed polymer sample might be selectively retained within the column, which then may lead to an error in the determination of the critical point. Consequently, the determined molar masses and molar mass distributions values of polymers obtained from two-dimensional chromatography by coupling LCCC with SEC (LCCC × SEC) will be erroneous also for that part of macromolecules which is otherwise separated in the SEC mode [4,32].

Although polyolefins have been industrially produced for 70 years, LCCC of polyethylene (PE) or polypropylene (PP) could not have been realized until recently, because experimental conditions for their reversible adsorption were not known. The breakthrough came with the application of porous graphitic carbon (PGC) as stationary phase material i.e., when Macko et al. described the separation of PE, PP and ethylene/1-alkene copolymers using porous graphitic carbon (PGC) [33,34]. This has enabled to realize LCCC for linear polyethylene (PE) by Mekap et al. [25] and of PP by Bhati et al. [24] using the SEC-LAC method as well as the method according to Cools and Bashir.

Commercially available polyolefins regularly contain high or even ultra-high molar mass fractions, which may be eventually adsorbed at critical conditions. Therefore, a new method for determination the critical point will be described in this paper, which is based on monitoring the recovery of a single polymer sample with high average molar mass from a column. This method will be tested to determine the LCCC of Isotactic Polypropylene (*it*-PP), Syndiotactic Polypropylene (*st*-PP) and PE.

2. Experimental

2.1. Polymer samples

An *it*-PP standard (catalogue number: PP350K) was obtained from American Polymer Standards Corporation (Mentor, Ohio, USA). An *st*-PP standard (catalogue number: 452149) was obtained from Sigma-Aldrich (Milwaukee, USA). A linear PE standard (catalogue number: PSS-pe181k) was obtained from Polymer Standards Service, Mainz, Germany. The characteristics of the polymer samples used are shown in Table 1. The data about the weight average molar mass (M_W) and the dispersity index (Đ) of these samples were provided by the suppliers.

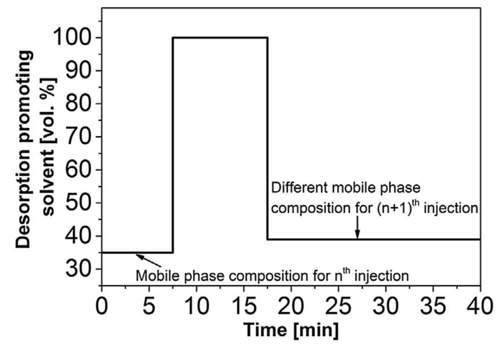

Fig. 1. Composition of the mobile phase for nth and (n + 1)th injection versus time.

The polymer standards were dissolved (~2–3 mg/mL) at 160 °C for ~2 h in a pure adsorption promoting solvent. The adsorption promoting solvent is usually a thermodynamically poor solvent for the polymer of interest, which promotes it's adsorption on the stationary phase.

2.2. Mobile phase and stationary phase

1,2-Dichlorobenzene (ODCB), 1,2,4-trichlorobenzene (TCB), 2-octanol and 2-ethyl-1-hexanol (2E1H) were obtained from Merck KGaA, Darmstadt, Germany. ODCB and TCB were distilled prior to use, the remaining mobile phases were used as received. A Hypercarb^{TM} column 4.6 (mm) × 250 (mm) I.D. × L. containing porous graphitic carbon with an average particle diameter of 5 µm, a surface area of 122 m^2/g and a pore diameter of 250 Å was purchased from ThermoFisher Scientific, Dreieich, Germany.

2.3. High temperature high performance liquid chromatography

A high temperature liquid chromatograph (PL-XTR 220, Polymer Laboratories, Church Stretton, England) was used for the experiments. The temperatures of the sample oven, transfer line, injector and the column oven were set on 160 °C. The sample loop had a volume of 50 µL. An evaporative light scattering detector (ELSD, model PL-ELS 1000, Polymer Laboratories, Church Stretton, England) was used to monitor the composition of the effluent. Temperatures of the nebulizer and evaporator were set to 160 °C and 260 °C, respectively, whereas a nitrogen flow rate of 1.5 L/min was used for nebulization of the effluent. Data collection (1 point/s) and processing were done with WinGPC-software (Polymer Standards Service, Mainz, Germany).

A high-pressure binary gradient pump (Agilent, Heilbronn, Germany) was used for mixing and pumping of the binary mobile phases. The mobile phase flow rate was maintained constant at 1 mL/min. Total delay volume of the system was determined according to Ginzburg et al. [35]. The delay volume of the system was 5.1 mL.

The following programme was used for pumping of the mobile phase (Fig. 1).

The polymer was injected into such a mobile phase composition (first step in Fig. 1) that it was fully or partially adsorbed in the column. Subsequently, the retained polymer was desorbed and eluted after pumping a desorption promoting solvent i.e., TCB or ODCB (second step in Fig. 1). For the next injection the column was purged with the new mobile phase (final step in Fig. 1), which contained an increased concentration of the desorption promoting solvent. This procedure was repeated until no peak of the polymer appeared on the chromatogram after purging the column with TCB or ODCB.

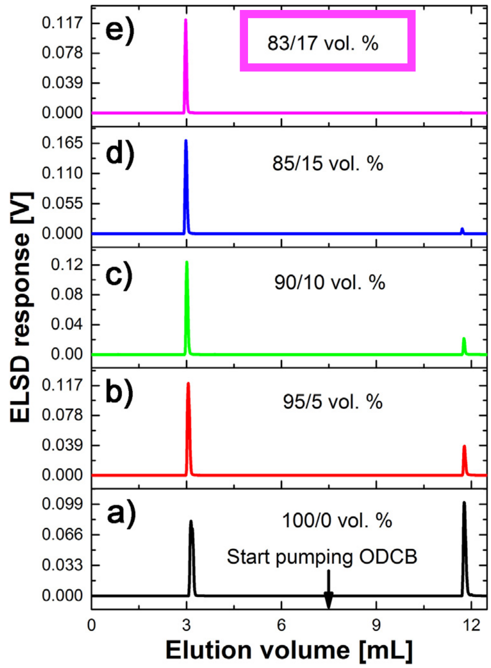

Fig. 2. Stack plot with elugrams in different mobile phase compositions for *it*-PP^{348}. Mobile phase: 2-octanol/ODCB. Composition of the mobile phase is shown in the figure in vol.%.

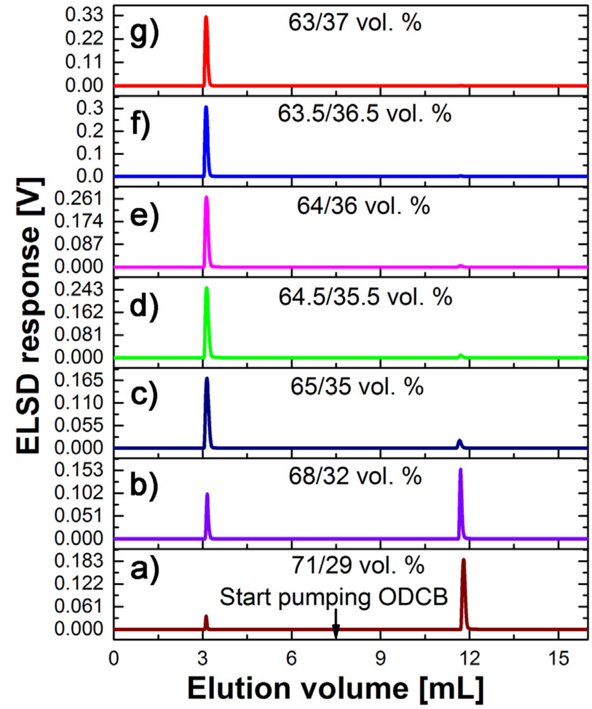

Fig. 3. Stack plot with elugrams in different mobile phase compositions for *st*-PP^{196}. Mobile phase: 2-octanol/ODCB. Composition of the mobile phase is shown in the figure in vol.%.

An isocratic pump can be used for the realization of the step gradient (Fig. 1), by preparing the needed mobile phase composition in a bottle and pumping it into the column, however, use of a gradient pump eliminates manual preparation of the requested binary mobile phases.

3. Results and discussions

3.1. Determination of the critical conditions for isotactic polypropylene

it-PP^{348} was dissolved in an adsorption promoting solvent (2-octanol) and then the solution was injected into the column flushed with a binary mobile phase (2-octanol/ODCB) at a particular composition (with a higher percentage of adsorption promoting solvent). In the subsequent injection the composition of the mobile phase was changed accordingly to reduce the adsorption of the polymer in the Hypercarb^{TM} column. This process was repeated until no fraction of the polymer was adsorbed on the stationary phase. The elugrams for selected injections are shown in Fig. 2.

A part of the *it*-PP^{348} sample eluted from the column (Fig. 2a), while another part of the sample was adsorbed in the column. The polymers eluted in a mixed mode between LCCC and the adsorption mode, which means that macromolecules with lower molar mass were poorly adsorbed. Fig. 2 indicates that they eluted near to critical conditions. As consequence, their SEC separation was not observed (the first peak in Fig. 2a). The retained portion (macromolecules with higher molar mass) was then desorbed by purging pure ODCB into the column – leading to the appearance of the second peak of *it*-PP^{348} on the chromatogram (Fig. 2a). The composition of the binary mobile phase was then changed, *it*-PP^{348} was again injected and the adsorbed part of the sample was eluted after pumping pure ODCB. From Fig. 2b–d it can be inferred that the increase in the concentration of the desorption promoting solvent leads to an increase in the intensity of the first peak, while the peak eluting after pumping pure ODCB decreases in intensity. Finally, at the critical conditions (i.e., 83/17 vol.% of 2-octanol/ODCB), the sample is eluted in a single peak (Fig. 2e). *it*-PP^{348} will elute in SEC mode at higher concentration of the desorption promoting solvent i.e., the first composition of the mobile phase, at which the polymer is fully eluted, corresponds to the critical conditions (Fig. 2e).

Fig. 2 illustrates the practical protocol i.e., sequence of injections and how many compositions of the mobile phase were used. Other details are described in the experimental part. It is usually recommended that for a successful elution under critical conditions the sample solvent composition should be similar to the mobile phase composition. Otherwise the elution may be prone to various chromatographic artefacts e.g., peak splitting, sample breakthrough [36] and limited recovery [4]. We emphasize that although the polymer samples were dissolved in the adsorption promoting solvent, the mentioned effects (peak splitting, sample breakthrough, limited recovery) were not observed. It is a fact that the experiments are done mainly not at the critical conditions, but in the adsorption mode. The elution in the adsorption mode is a function of the molar mass, hence a part of the polymer sample (with lower molar mass) eluted first and a part of the sample (with larger molar mass) was retained and eluted later (Fig. 2). Macromolecules with smaller molar mass eluted near the critical conditions and in such case the molar mass influence is eliminated or negligible.

3.2. Determination of the critical conditions for syndiotactic polypropylene

In general, *st*-PP is stronger adsorbed on the porous graphitic carbon surface compared to *it*-PP and thus it requires a larger concentration of the desorption promoting solvent (TCB or ODCB) to desorb the adsorbed *st*-PP from the graphitic surface [37]. The stack plot shown in Fig. 3 illustrates the identification of critical conditions for *st*-PP in the mobile phase 2-octanol/ODCB using the new method.

The LCCC for *st*-PP is identified as 63/37 vol.% of 2-octanol/ODCB (Fig. 3g).

Table 2
Composition of mobile phases under the critical conditions for *it*-PP, *st*-PP or PE. Column: Hypercarb™. Temperature: 160 °C.

Polymer and mobile phase	SEC-LAC plot [vol.%]	Bashir's method [vol.%]	Cools' method [vol.%]	New method [vol.%]
it-PP 2-octanol/ODCB	83/17	81/19	80.5/19.5	83/17
st-PP 2-octanol/ODCB	65/35	64.2/35.8	63.5/36.5	63/37
PE 2-octanol/ODCB	21/79	21.5/78.5	–	20.5/79.5
it-PP 2E1H/TCB	88/12	87.9/12.1	86.5/13.5	88/12
st-PP 2E1H/ODCB	65/35	66.8/33.2	–	64/36
PE *n*-decane/TCB	51.5/48.5	50.8/49.2	52/48	52/48

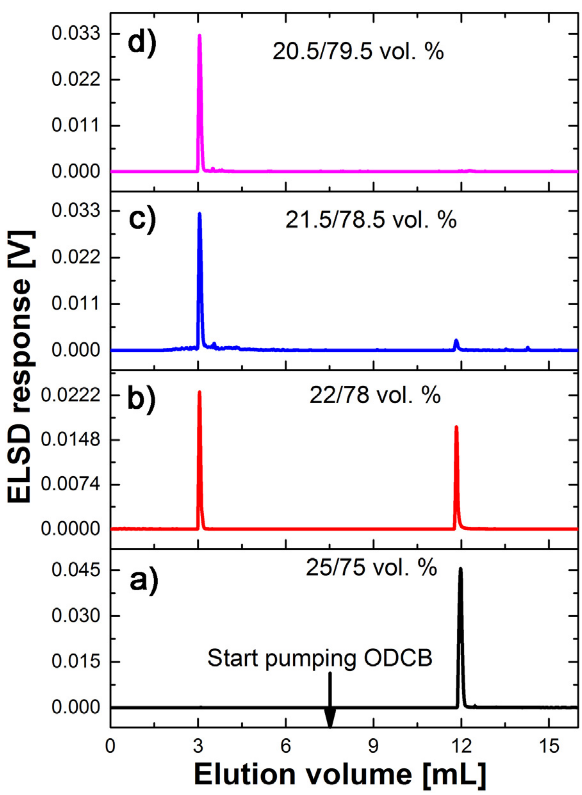

Fig. 4. Stack plot with elugrams in different mobile phase compositions for PE[181]. Mobile phase: 2-octanol/ODCB. Composition of the mobile phase is shown in the figure in vol.%.

3.3. Determination of the critical conditions for polyethylene

LCCC of PE was determined by using the sample PE[181]. The sample was injected into the Hypercarb™ column at different solvent compositions of 2-octanol and ODCB. After injecting the polymer at a particular composition, 100% desorption promoting solvent ODCB was pumped into the column with the aim to desorb and elute the polymer (Fig. 1). The solvent composition at which PE[181] fully eluted as a single peak and no portion of it eluted after pumping pure ODCB into the column corresponds to the critical conditions for PE in the mobile phase 2-octanol/ODCB at 160 °C. An overlay of the chromatograms for PE[181] at different ratios of 2-octanol/ODCB is shown in Fig. 4.

The LCCC for PE is identified as 20.5/79.5 vol.% of 2-octanol/ODCB (Fig. 4d).

The critical conditions for *it*-PP, *st*-PP and PE were determined in different solvent pairs [24,25] using the three methods described in the introduction and the results were compared with those obtained using the new method in Table 2.

The critical composition of the mobile phase for *it*-PP in 2-octanol/ODCB as well as in 2E1H/TCB obtained with the new method are identical with the critical composition determined using the SEC-LAC plot method. The same is valid for critical composition of the mobile phase for PE in *n*-decane/TCB obtained with the new method and Cools' method. Good agreement in the critical composition of the mobile phase for st-PP in 2-octanol/ODCB is found between the results obtained with the new method and Cools' method.

The maximum deviations in the critical compositions of the mobile phases are observed between Bashir's method and the new method (Table 2). This can be explained by the fact that the value of the critical composition is determined from E_{pmax} of a single polymer peak when using Bashir's method. For this purpose it is usually expected that the shape of the solvent gradient in a detector is proportionate to the shape of the gradient in the pump, being delayed by the volume (time) the gradient requires to reach the detector. It is known that the shape of the solvent gradient in the detector is not always parallel with the shape of the gradient generated in the pump. The reasons for this are the change in the mobile phase composition in the gradient and the effects of pore diffusion as well as the differing viscosity of constituents of the gradient [38]. As a result, the supposed critical composition of the mobile phase corresponding to the top of the polymer peak may vary from the real LCCC. The SEC-LAC plot method as well as Cool's method use elution volumes of series of polymer standards measured at isocratic conditions and thus gives more reliable values of the critical compositions of mobile phases (Table 2) compared to Bashir's method (only one polymer sample used in a solvent gradient). The new method utilizes isocratic mobile phases, which in turn makes it possible to use a refractive index detector instead of ELSD. Many chromatographic systems use non-solvent as one of the components of mobile phase for polymer separations. Typical example is hexane-tetrahydrofuran mixture for separation of polystyrenes, polyacrylates and many other polymers with LCCC [3,20,27,28]. Such mixture can also be used as an isocratic mobile phase according to the method described in this paper for the determination of critical conditions. Although, the polymer of interest should be fully soluble in all tested solvent/non-solvent mixtures.

It is expected that for symmetrical peaks or narrow distributed polymers with relatively lower molar masses the deviations in the critical conditions should be less compared to the asymmetrical peaks or broad distributed polymers with high or ultra-high molar masses. We summarize that a solvent gradient (as required with Bashir's method), a series of polymer samples (as required with SEC-LAC plot method and Cools' method), defined shape of peaks (symmetrical or asymmetrical) or precise values of elution volumes (as required with all three methods) are not a prerequisite of the new method, which apparently has a positive effect on the reliability of obtained results. This means that also a polymer sample with a broad or even bimodal molar mass distribution can be used for the measurements. Moreover, the recovery of the polymer samples of interest in LCCC is controlled, which is particularly important for commercial olefin copolymers, which regularly contain high molar mass fractions.

4. Conclusions

In this paper a new method for determination of the critical conditions was described and experimentally tested. The LCCC for polymers was identified using only one high molar mass polymer sample. The polymer sample was injected in different mobile phase compositions in LAC mode. After elution of the polymer from the column, sample's recovery from the column was checked by pumping pure desorption promoting solvent. The first composition of the mobile phase, when the polymer eluted with full recovery (i.e., no peak of polymer appeared on chromatogram after pumping pure desorption promoting solvent) corresponds to the critical conditions. Limited recovery of analyzed polymers is usually not desirable in LCCC. Namely, limited sample recovery may lead to serious errors while applying the LCCC to selectively separate end-functionalized polymers, polymers with different tacticity or architecture, block copolymers of A–B or A–B–A type and polymer blends. Other advantage of this method is that the LCCC of a polymer can be realized with a single high molar mass sample and an isocratic pump, if a gradient pump is not available. The results obtained with the new method were compared with the results obtained with the published methods and they were found in good agreement.

Determination of the critical conditions for polyolefins will enable analytical separation of end-functionalized polyethylene and polypropylene as well as separation of diblock or triblock polyolefin copolymers. Synthesis of such polyolefins is in focus of many academic and industrial laboratories. The proposed method for the determination of the LCCC has potential to be applied not only for polyolefins but also for other polymers.

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